PHYSICS

1) The radius of a circular plate increases at the rate of 0.1 cm/sec. At what rate does the area increase when the radius of plate is $\frac{5}{11}$ cm ?

(1) 1 cm^2/sec
(2) 0.1 cm^2/sec
(3) 0.6 cm^2/sec
(4) 0.29 cm^2/sec

2) The value of sin31° is :-

(1) $\frac{1}{2}$
(2) $\frac{1}{2}\left(1+\frac{\pi\sqrt{3}}{180}\right)$
(3) $\frac{1}{2}\left(\sqrt{3}+\frac{\pi}{180}\right)$
(4) None

3) If $\vec{P}=\vec{F}_1+\vec{F}_2$ and $\vec{Q}=\vec{F}_1-\vec{F}_2$; the magnitude of $\vec{F}_1$ and $\vec{F}_2$ are $3\sqrt{2}N$ and $4\sqrt{2}N$ respectively, then the value of $\sqrt{|\vec{P}|^2+|\vec{Q}|^2}$ is :-

(1) 5N
(2) $6\sqrt{2}N$
(3) 10N
(4) $7\sqrt{2}N$

4) If vector $2\hat{i}+2\hat{j}-2\hat{k}$, $-\hat{i}+2\hat{j}+2\hat{k}$ and $5\hat{i}+y\hat{j}+\hat{k}$ are coplanar, the value of y is :-

(1) 81
(2) 36
(3) 23
(4) 9

5) The angle subtended by the vector $\vec{A}=4\hat{i}+3\hat{j}+12\hat{k}$ with the x-axis is :-

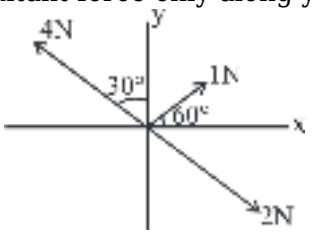
(1) $\sin^{-1}\left(\frac{3}{13}\right)$

(2) $\sin^{-1}\left(\frac{4}{13}\right)$

(3) $\cos^{-1}\left(\frac{4}{13}\right)$

(4) $\cos^{-1}\left(\frac{3}{13}\right)$

6) Three forces acting on a body are shown in figure. To have the resultant force only along y-



direction, the magnitude of the minimum additional force needed is :-

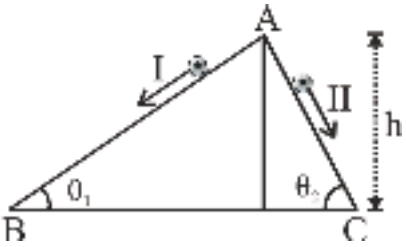
(1) 0.5N

(2) 1.5N

(3) $\frac{\sqrt{3}}{4}$N

(4) $\sqrt{3}$N

7) Two inclined frictionless tracks, one gradual and the other steep, meet at A from where two stones are allowed to slide down from rest, one on each track as shown in figure. Which of the



following statement is correct?

(1) Both the stones reach the bottom at the same time but not with the same speed.

(2) Both the stones reach the bottom with the same speed and stone I reaches the bottom earlier than stone II.

(3) Both the stones reach the bottom with the same speed and stone II reaches the bottom earlier than stone I.

(4) Both the stones reach the bottom at different times and with different speeds.

8) A bullet moving with a velocity of 200 cm/s penetrates a wooden block and comes to rest after traversing 4 cm inside it. What velocity is needed for travelling of 9 cm in same block ?

(1) 100 cm/s

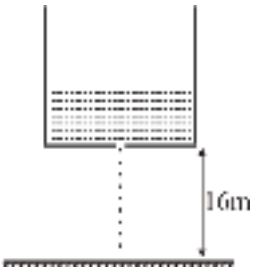
(2) 136.2 cm/s

(3) 300 cm/s

(4) 250 cm/s

9) Water drops fall with negligible velocity at regular intervals from a hole at the bottom of a vessel placed 16 m high from the ground. The ninth drop is about to fall when the first drop just falls on the

floor. Find the distance between the third and fifth drop at this instant, in meters.



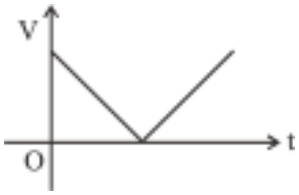
(1) 2

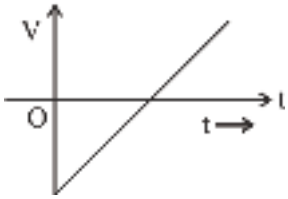
(2) 5

(3) 8

(4) 9

10) A particle moves along x-axis in such a way that its x-coordinates varies with time t according to equation $x = 4 - 2t + t^2$. The speed of the particle will vary with time as:

(1) 

(2) 

(3) 

(4) 

11)

Which of the following quantities remain constant during projectile motion?

(A) Average velocity between two points.

(B) Average speed between two points

(C) $\frac{d\vec{v}}{dt}$

(D) $\frac{d^2}{dt^2}(\vec{v})$

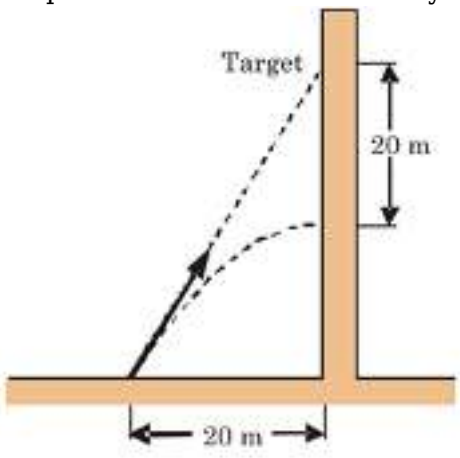
(1) A only

(2) C only

(3) A and C

(4) C and D

12) A shooter fires his gun by aiming on a target but the bullet hits the wall horizontally 20 m below the target. The bullet is fired from a point that is 20 m away from the wall. The horizontal component of the bullet's velocity is :



(1) 10 m/s

(2) 15 m/s

(3) 20 m/s

(4) 25 m/s

13) Two trains one of length 100 m and another of length 125 m, are moving in mutually opposite directions along parallel lines. They meet each other when each train has speed of 10 m/s. If their acceleration are 0.3 m/s^2 and 0.2 m/s^2 respectively, then the time they take to pass each other will be :-

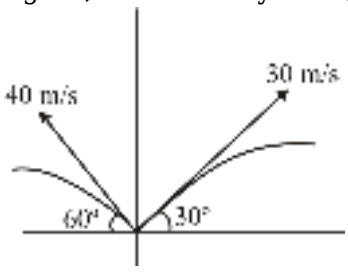
(1) 5 s

(2) 10 s

(3) 15 s

(4) 20 s

14) Two particles A and B are thrown from same point and in same vertical plane, as shown in figure, with velocity 30 m/s & 40 m/s respectively. Find the separation between them after 2 sec.



(1) 100 m

(2) 115 m

(3) 150 m

(4) 75 m

15) A man standing in the rain holds an umbrella at an angle 30° from vertical. When he throws the umbrella and start running with speed 10 km/hr, he finds that rain drops are hitting his head vertically. Find the actual speed of rain drops :-

(1) 20 km/hr

(2) $10\sqrt{3}$ km/hr

(3) $\frac{10}{\sqrt{3}}$ km/hr

(4) $20\sqrt{3}$ km/hr

16) A boat-man can row a boat to make it move with a speed of 20 km/h in still water. River flows steadily at the rate of 10 km/h and the width of the river is 2 km. If the boat man cross the river along the minimum distance of approach, then time elapsed in rowing the boat will be :–

(1) $\frac{2}{5\sqrt{3}}$ h

(2) $\frac{2\sqrt{3}}{5}$ h

(3) $\frac{1}{5\sqrt{3}}$ h

(4) $\frac{5\sqrt{2}}{3}$ h

17) **Assertion (A)**:- A coin is allowed to fall in a train moving with constant velocity. Its trajectory is a straight line as seen by an observer attached to the train.
Reason (R) :- An observer on ground will see the path of coin as a parabola.

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(2) **(A)** is correct but **(R)** is not correct

(3) **(A)** is incorrect but **(R)** is correct

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

18) Length of rectangular box is measured using vernier callipers whose 20 divisions of vernier scale coincides with 19 divisions of main scale. If zero of vernier scale lies between 5.1 cm and 5.3 cm reading of main scale while measuring length, and it has been observed that 6th division of vernier scale coincides with a main scale division, then length of box will be :-

(1) 5.16 mm

(2) 5.16 cm

(3) 5.36 cm

(4) 5.36 mm

19) Measure of two quantities along with the possible errors in respective measuring instrument is:
A = (2.5 ± 0.5)m/sec, B = (0.10 ± 0.01)s
The value of AB will be

(1) (0.25 ± 0.08) m

(2) (0.25 ± 0.5) m

(3) (0.25 ± 0.05) m

(4) (0.25 ± 0.135) m

20) Which of the following sets have different dimensions ?

(1) Pressure, Young's modulus, stress

(2) Emf, potential difference, electric potential

(3) Heat, work done, energy

(4) Dipole moment, electric flux, electric field

21) If $F = \frac{v}{C \ln(xb)}$, then :-

(1) F and v denote force and velocity, the dimensions of C are [MT]

(2) x denote distance, the dimensions of b are $[L^{-1}]$

(3) the dimensions of $\frac{v}{C}$ can never be same as F

(4) the dimensions of x must be same as $\frac{v}{Cb}$

22)

The speed (ν) of ripples on the surface of water depends on surface tension (σ), density (δ) and wavelength (λ). The square of speed (ν) is proportional to :

(1) $\frac{\sigma}{\delta\lambda}$

(2) $\frac{\delta}{\sigma\lambda}$

(3) $\frac{\lambda}{\sigma\delta}$

(4) δλσ

23) Two numbers a = 0.92 and b = 0.08 are given. The number of significant figure present in result after the following operations a + b, a – b, a × b and a/b respectively are :-

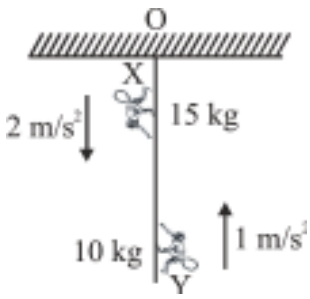
(1) 2, 2, 2, 2

(2) 3, 3, 2, 2

(3) 3, 2, 1, 1

(4) 3, 2, 2, 2

24) According to given figure monkey X of mass 15 kg moving downwards with acceleration 2 m/s^2 and monkey Y of mass 10 kg is moving upwards with acceleration 1 m/sec^2 on light rope then find



out force exerted on point O :–

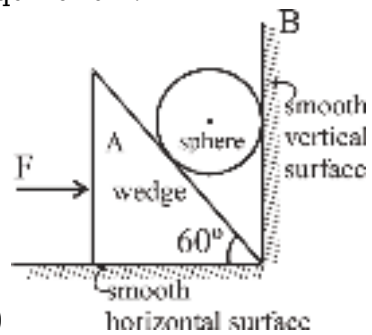
(1) 210 N

(2) 230 N

(3) 21 N

(4) 200 N

25) The 5 kg homogeneous smooth sphere rests on inclined surface of triangular wedge A and bears against the smooth vertical wall B. Find the horizontal force 'F' required to be applied on wedge to hold the system in equilibrium.



(Take g = 10 m/sec^2)

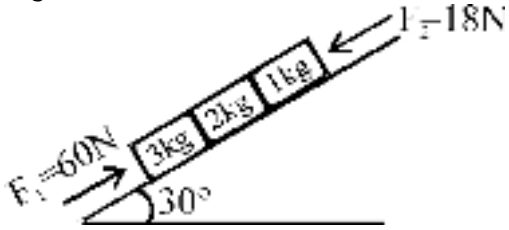
(1) 100 N

(2) 50 N

(3) $50\sqrt{3}$N

(4) $100\sqrt{3}$N

26) In the situation shown in figure all surfaces are smooth then acceleration of 2 kg block is :–



(1) 2 m/s^2 downwards

(2) 2 m/s^2 upward

(3) 5 m/s^2 downward

(4) 5 m/s^2 upward

27) Two blocks A and B of masses 4 m and m respectively are connected by a massless and inextensible string. The whole system is suspended by a massless spring as shown in figure. The



magnitudes of acceleration of A & B immediately after the string is cut, are respectively:-

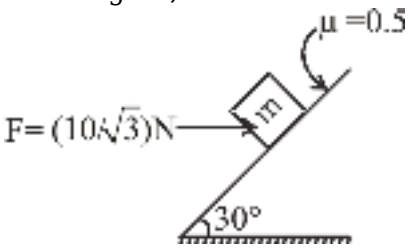
(1) g, g

(2) $\frac{g}{4}, \frac{g}{4}$

(3) $g, \frac{g}{4}$

(4) $\frac{g}{4}, g$

28) If the mass of block is 1 kg and a force of $10/\sqrt{3}N$ is applied horizontally on the block as shown in the figure, then the frictional force acting on the block is :- ($g = 10\ ms^2$)



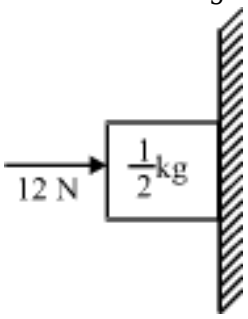
(1) zero

(2) $\frac{10}{\sqrt{3}}N$

(3) $\frac{20}{\sqrt{3}}N$

(4) 5 N

29) A horizontal force 12 N pushes a block weighing 1/2 kg against a vertical wall. The coefficient of static friction between the wall and the block is 0.5 and the coefficient of kinetic friction is 0.35. Assuming that the block is not moving initially. Which one of the following choices is correct :- (Take



$g = 10\ m/s^2$)

(1) Block moves vertically downwards

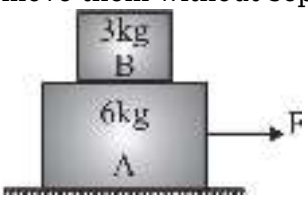
(2) Block moves vertically upwards

(3) Block will not move and force exerted on the block by the wall is 12 N

(4) Block will not move and force exerted on the block by the wall is 13 N

30) Two blocks A and B of masses 6 kg and 3 kg rest on a smooth horizontal surface as shown in the

figure. If coefficient of friction between A and B is 0.4, the maximum horizontal force which can move them without separation is :-



(1) 72 N

(2) 40 N

(3) 36 N

(4) 20 N

31) When an object is placed at a distance of 25 cm from a mirror, the magnification is m_1. The object is moved 15 cm away with respect to the earlier position, magnification becomes m_2. If $m_1/m_2 = 4$, the focal length of the mirror is :

(1) 10 cm

(2) 30 cm

(3) 15 cm

(4) 20 cm

32) A ray of light is incident on a surface of glass slab at an angle of 45°. If the lateral shift produced per unit thickness is $\frac{1}{\sqrt{3}}$ m, the angle of refraction is :-

(1) $\tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$

(2) $\tan^{-1}\left(1-\sqrt{\frac{2}{3}}\right)$

(3) $\sin^{-1}\left(1-\sqrt{\frac{2}{3}}\right)$

(4) $\tan^{-1}\left(1-\sqrt{\frac{2}{\sqrt{3}-1}}\right)$

33) A ray of monochromatic light is incident on one refracting face of a prism of angle 75°. It passes through the prism and is incident on the other face at the critical angle. If the refractive index of the material of the prism is $\sqrt{2}$, the angle of incidence on the first face of the prism is:-

(1) 30°

(2) 45°

(3) 60°

(4) 0°

34) If f_1 and f_2 represent the first and second focal lengths of a single refracting surface, then

(assuming cartesian sign convention) :-

(1) $f_2 + f_1 = 0$

(2) $f_2 + \mu f_1 = 0$

(3) $f_1 + \mu f_2 = 0$

(4) $f_1 f_2 = 1$

35) **Assertion :** Focal length of a lens depends on the wavelength of light used.
Reason : The more the wavelength, less is the focal length.

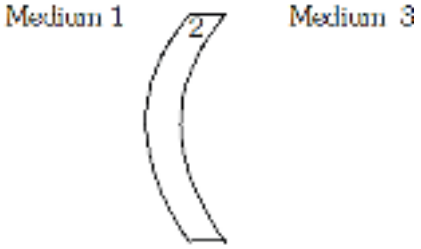
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

36) The focal length of the lens is nR as shown in the figure. If the radii of curvature of both the surfaces are equal to R, then value of n is : ($\mu_1 = 1$, $\mu_2 = 0.5$ and $\mu_3 = 2$)



(1) 1

(2) 2

(3) 1.5

(4) 2.5

37) When an object is at a distance x and y from the optical centre of a lens, a real and a virtual images are formed, respectively with same magnification. The focal length of lens is–

(1) $x + y$

(2) $\frac{(x+y)}{2}$

(3) $x + \frac{y}{2}$

(4) $\sqrt{xy}$

38) An astronomical refracting telescope is being used by an observer to observe planets in normal adjustment. The focal lengths of the objective and eye piece used in the construction of the telescope are 20 m and 2 cm respectively. Consider the following statements about the telescope :
(a) The distance between the objective and eye piece is 20.02 m
(b) The magnification of the telescope is (–) 1000
(c) The image of the planet is erect and diminished
(d) The aperture of eye piece is smaller than that of objective
The correct statements are :

(1) (a), (b) and (c)

(2) (b), (c) and (d)

(3) (c), (d) and (a)

(4) (a), (b) and (d)

39) Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : For a simple microscope, the angular size of the object equals the angular size of the image.

Reason R : Magnification is achieved as the small object can be kept much closer to the eye than 25 cm and hence it subtends a large angle.

In the light of the above statements, choose the most appropriate answer from the options given below :

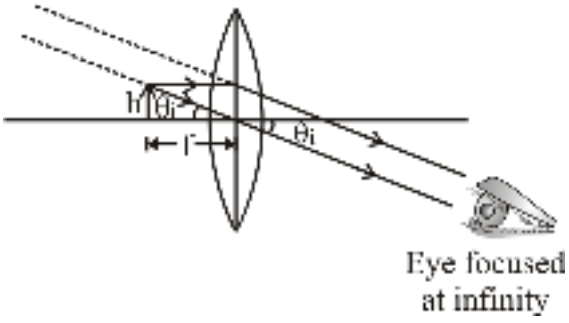
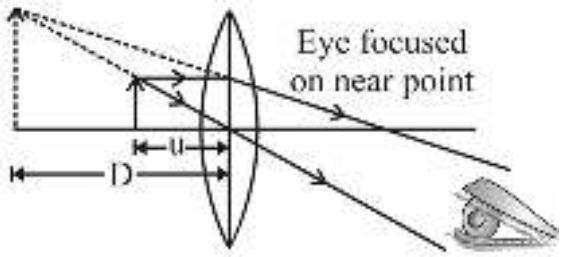
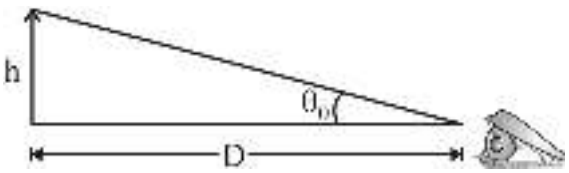
(1) A is true but R is false

(2) Both A and R are true but R is NOT the correct explanation of A.

(3) Both A and R are true and R is the correct explanation of A.

(4) A is false but R is true.

40) Match the Column I (different cases of simple microscope) with Column II (ray diagram) and select the correct answer from the codes given below.

	Column I		**Column II**
A	The magnifying lens is located such that image is at the near point.	1.	
B.	The angle substended by the object, is the same as that at the near point.	2.	
C.	If the object is near the focal point of the lens, the image is formed far off but closer than infinity.	3.	

	A	B	C
(1)	1	2	3
(2)	3	1	2
(3)	1	3	2
(4)	2	3	1

(1) 1

(2) 2

(3) 3

(4) 4

41) In YDSE, d = 1 mm, λ = 6000 Å and D = 1 m. The minimum distance between two points on screen having 75% intensity of the maximum intensity, will be :-

(1) 0.50 mm

(2) 0.40 mm

(3) 0.30 mm

(4) 0.20 mm

42) In the Young's double slit experiment, the ratio of intensities of bright and dark fringes is 9. This means that :-
(A) The intensities of two slits are 5 and 4 units respectively.
(B) The intensities of two slits are 4 and 1 units respectively.
(C) The ratio of their amplitudes is 3.
(D) The ratio of their amplitudes is 2.

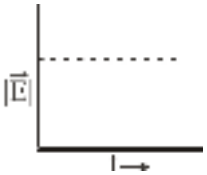
(1) B, D

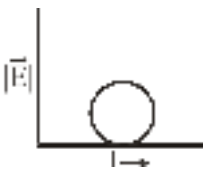
(2) B, C

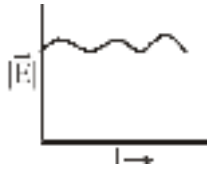
(3) A, C

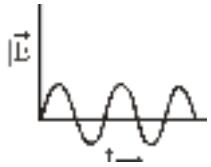
(4) A, D

43) Which of the following diagrams represent the variation of electric field vector with time for a circularly polarized light ?

(1) 

(2) 

(3) 

(4) 

44) In a double slit experiment, the two slits are 1 mm apart and the screen is placed 1 m away. A monochromatic light of wavelength 500 nm is used. What will be the width of each slit for obtaining ten maxima of double slit within the central maxima of single slit pattern ?

(1) 0.1 mm

(2) 0.5 mm

(3) 0.02 mm

(4) 0.2 mm

45) What is the ratio of slit width when amplitude of light waves have a ratio 9 : $\sqrt{5}$?

(1) $\frac{3}{\sqrt[4]{5}}$

(2) $\frac{81}{\sqrt[3]{5}}$

(3) $\frac{3}{\sqrt[3]{5}}$

(4) $\frac{81}{5}$

CHEMISTRY

1) Which may be a reversible reaction :-

(1) $H_2 + I_2 \rightarrow 2HI$

(2) $H_2SO_4 + Ba(OH)_2 \rightarrow BaSO_4\downarrow + 2H_2O$

(3) $NaCl + AgNO_3 \rightarrow NaNO_3 + AgCl\downarrow$

(4) $2KClO_3 \rightarrow 2KCl + 3O_2\uparrow$

2) Select the endothermic reaction :

(1) $2H_2 + O_2 \longrightarrow 2H_2O$

(2) $N_2 + O_2 \longrightarrow 2NO$

(3) $2NaOH + H_2SO_4 \longrightarrow Na_2SO_4 + 2H_2O$

(4) $3O_2 + C_2H_5OH \longrightarrow 2CO_2 + 3H_2O$

3) The equilibrium constant of a reaction is 20.0. At equilibrium, the rate constant of forward reaction is 10. The rate constant for backward reaction is :

(1) 0.5

(2) 2.0

(3) 10.0

(4) 200.0

4)

What is observed at equilibrium point in the reaction?
$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

(1) Equal volumes of PCl_5, PCl_3 and Cl_2 are present

(2) Equal masses of PCl_5, PCl_3 and Cl_2 are present

(3) The concentrations of PCl_5, PCl_3 and Cl_2 become constant

(4) Reaction will stop

5) For the reaction $A_2(g) + 2B_2(g) \rightleftharpoons 2C_2(g)$. The partial pressure of A_2 and B_2 at equilibrium are 0.8 and 0.4 atm respectively and total pressure is 2.8 atm then the K_p is

(1) 20

(2) 5

(3) 0.02

(4) 0.2

6) Four vessels of 10 L each A, B, C and D contain 16 g CH_4, 18 g of H_2O(□), 71 g of Cl_2 and 44 g of CO_2 respectively. Which has different active mass ?

(1) A

(2) B

(3) C

(4) D

7) For the reaction X + Y ⇌ XY at equilibrium if the concentration of X and Y are doubled, the equilibrium constant

(1) is doubled

(2) is reduced to half

(3) remains unaltered

(4) is tripled

8) An equilibrium mixture of reaction
$2H_2S_{(g)} \rightleftharpoons 2H_{2(g)} + S_{2(g)}$ has 0.5 mole H_2S, 0.10 mole H_2 and 0.4 mole S_2 in one litre vessel. The value of equilibrium const (K) in mol L^{-1} is

(1) 0.004

(2) 0.008

(3) 0.016

(4) 0.160

9) Partial pressure of O_2 in the reaction at equilibrium is,
$2Ag_2O(s) \rightleftharpoons 4Ag(s) + O_2(g)$

(1) K_p

(2) $\sqrt{K_p}$

(3) $\sqrt[3]{K_p}$

(4) $2K_p$

10) $CuSO_4 \cdot 5H_2O(s) \rightleftharpoons CuSO_4 \cdot 3H_2O(s) + 2H_2O(g)$; $Kp = 4 \times 10^{-4}$ atm^2. The vapour pressure of water at equilibrium will be :-

(1) 0.2 atm

(2) 10 atm

(3) 0.02 atm

(4) 2 atm

11) For this reaction $PCl_3 + Cl_2 \rightleftharpoons PCl_5$, K_c is 26 the K_p will be (T = 250 °C)

(1) 0.605

(2) 0.95

(3) 0.83

(4) 0.46

12) For which reaction will $K_p = K_c$?

(1) $2CO(g) + O_2(g) \rightleftharpoons 2CO_2(g)$

(2) $H_2(g) + Br_2(g) \rightleftharpoons 2HBr(g)$

(3) $C(s) + 2H_2O(g) \rightleftharpoons 2H_2(g) + CO_2(g)$

(4) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

13) $X_2 + O_2 \rightleftharpoons 2XO$, $K_1 = 1.5 \times 10^{-4}$
$2XO_2 \rightleftharpoons X_2 + 2O_2$, $K_2 = 2 \times 10^4$
Which oxide is more stable :-

(1) XO

(2) XO_2

(3) both are equally stable

(4) None of these

14)

For the gaseous reversible reaction at 700 K,
$H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$
The value of an equilibrium constant is 57, then at equilibrium the system will contain :-

(1) Mostly products
(2) Mostly reactants
(3) Appreciable concentrations of both reactants and products
(4) Can't be predict

15)

$K_p \rightarrow 10^{-2} \quad 10^{-3}$
Temperature $\rightarrow$ 400K 450K

What would you consider by above information:-
(1) Equilibrium constant increases with increase in concentration
(2) more molecules form on left hand side
(3) Energy is released
(4) None

16) Value of equilibrium constant depend on -

(1) Temperature
(2) Volume
(3) Pressure
(4) Catalyst

17) For $N_2 + 3H_2 \rightleftharpoons 2NH_3$ equilibrium constant is K then equilibrium constant for
$2N_2 + 6H_2 \rightleftharpoons 4NH_3$ is

(1) $\sqrt{K}$
(2) K^2
(3) K/2
(4) $\sqrt{K+1}$

18) For the following three reactions I, II and III, equilibrium constants are given :– (I) $CO_{(g)} + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)$; K_1
(II) $CH_4(g) + H_2O(g) \rightleftharpoons CO(g) + 3H_2(g)$; K_2
(III) $CH_4(g) + 2H_2O(g) \rightleftharpoons CO_2(g) + 4H_2(g)$; K_3
Which of the following relations is correct ?

(1) $K_1\sqrt{K_2} = K_3$
(2) $K_2K_3 = K_1$
(3) $K_3 = K_1K_2$
(4) $K_3 = K_2^2 \cdot K_1$

19) 4 mol of PCl_5 are heated at constant temperature in closed container. If degree of dissociation

(α) for PCl_5 is 0.5 then calculate total number of moles at equilibrium :-

(1) 4.5

(2) 6

(3) 3

(4) 4

20) At a certain temperature only 25% HI dissociated according to reaction in 1L flask.
$2HI(g) \rightleftharpoons H_2(g) + I_2(g)$
If volume of flask increases to 2L. then :

(1) α increases

(2) α decreases

(3) α remain same

(4) Data insufficient

21) The equilibrium constant (K_P) for the reaction,
$PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$ is 16.
If the volume of the container is reduced to one-half its original volume, the value of K_P for the reaction at the same temperature will be :-

(1) 32

(2) 64

(3) 16

(4) 4

22) $AB_3(g)$ is dissociates as $AB_3(g) \rightleftharpoons AB_2(g) + \frac{1}{2}B_2(g)$. When the initial pressure of AB_3 is 800 torr and the total pressure developed at equilibrium is 900 torr. What percentage of $AB_3(g)$ is dissociated?

(1) 10%

(2) 20%

(3) 25%

(4) 30%

23) In the following reaction : $2NO(g) + Cl_2(g) \rightleftharpoons 2NOCl(g)$ It is observed that equilibrium is not attained and the rate of forward reaction is greater than rate of backward reaction. Which of the following is true for the reaction ?

(1) $K_p = Q_p$

(2) $Q_p > K_p$

(3) $Q_p < K_p$

(4) $Q_p \geqslant K_p$

24) 0.6 mole of PCl_5, 0.3 mole of PCl_3 and 0.5 mole of Cl_2 are taken in a 1L flask to obtain the following equilibrium :

$PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$
If the equilibrium constant K_c for the reaction is 0.2 Predict the direction of the reaction :-

(1) Forward direction
(2) Backward direction
(3) Direction of the reaction cannot be predicted
(4) Reaction does not move in any direction

25) Given the following reaction at equilibrium
$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$. At constant pressure if some inert gas is added to the system. Then

(1) More $NH_{3(g)}$ is produced
(2) less $NH_{3(g)}$ is produced
(3) No affect on the equilibrium
(4) K_p of the reaction is decreased

26) For the equilibrium
$SO_{2(g)} + \frac{1}{2} O_{2(g)} \rightleftharpoons SO_{3(g)}$ $\Delta H = -99$ kJ
The extent of forward reaction can be increased by :-

(1) Removing SO_3
(2) Using a catalyst
(3) Raising the temperature
(4) Lowering the pressure

27) Match **List-I** with **List-II** :

List-I Equilibria		**List-II Conditions/Parameters**	
A.	Liquid ⇌ Vapour	I.	Melting point
B.	Solid ⇌ Liquid	II.	Boiling point
C.	Solid ⇌ Vapour	III.	Sublimation point
D.	Solute (Solid) ⇌ Solute (solution)	IV.	Saturated solution
		V.	Unsaturated solution

Choose the **correct** answer from the options given below :
(1) A–I, B–II, C–III, D–IV
(2) A–III, B–II, C–I, D–IV
(3) A–II, B–I, C–III, D–V
(4) A–II, B–I, C–III, D–IV

28) **Statement-I :** For the physical equilibrium $H_2O(s) \rightleftharpoons H_2O(\square)$ on increasing temperature and increasing pressure, more water will form.

Statement-II : Since forward reaction is endothermic in nature and volume of water is greater than that of volume of ice.

(1) Both **Statement I** and **Statement II** are incorrect.
(2) **Statement I** is correct but **Statement II** is incorrect.
(3) **Statement I** is incorrect but **Statement II** is correct.
(4) Both **Statement I** and **Statement II** are correct.

29)

On increasing temperature, the equilibrium constant of exothermic and endothermic reactions respectively :-

(1) Increases and Decreases
(2) Decreases and Increases
(3) Increases and Increases
(4) Decreases and Decreases

30) Le-Chatelier's principle is applicable only to a :

(1) system in equilibrium
(2) irreversible reaction
(3) homogeneous reaction
(4) heterogeneous reaction

31) At 250°C, the observed vapour density of PCl_5 is 75. The degree of dissociation of PCl_5 would be : [mol. wt. of PCl_5 = 210] :-

(1) 75 %
(2) 40 %
(3) 52.5 %
(4) 70 %

32) Vapour density of the equilibrium mixture of the reaction $2NH_3(g) \rightleftharpoons N_2(g) + 3H_2(g)$ is 6.0. Percent dissociation of ammonia gas is:

(1) 13.88
(2) 58.82
(3) 41.66
(4) 30

33) **Statement I :** Catalyst lowers the activation energy for the forward and backward reactions by exactly the same amount.
Statement II : Catalyst appear in the balanced chemical equation.

(1) Both statement I and statement II are correct.
(2) Statement I is correct but statement II is incorrect.

(3) Statement I is incorrect but statement II is correct.

(4) Both statement I and statement II are incorrect.

34)

Match the following columns :

	Column-I (Chemical reaction)		**Column-II (Relation between K_P and K_C)**
(i)	$C_{(s)} + CO_{2(g)} \rightleftharpoons 2CO_{(g)}$	(a)	$K_P = K_C$
(ii)	$N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$	(b)	$K_P = K_C(RT)$
(iii)	$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$	(c)	$K_P = K_C(RT)^{-1}$
(iv)	$2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)}$	(d)	$K_P = K_C(RT)^{-2}$

(1) (i)–(b), (ii)–(a), (iii)–(d), (iv)–(c)

(2) (i)–(d), (ii)–(a), (iii)–(b), (iv)–(c)

(3) (i)–(a), (ii)–(b), (iii)–(d), (iv)–(c)

(4) (i)–(c), (ii)–(a), (iii)–(d), (iv)–(b)

35) Given below are two statements :-
Statement-I : In reversible chemical reaction, equilibrium will establish when concentration of reactants and products are equal.
Statement-II : All the measurable properties of the system are variable at equilibrium state.
In the light of above statements, select the correct option :-

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct

(4) Both **Statement I** and **Statement II** are correct.

36) Match the columns :

Column-I		**Column-II**	
(A)	For the equilibrium $NH_4I(s) \rightleftharpoons NH_3(g)+HI(g)$, if pressure is increased at equilibrium	(p)	Forward shift
(B)	For the equilibrium $N_2(g)+3H_2(g) \rightleftharpoons 2NH_3(g)$ If volume is increased at equilibrium	(q)	No change

(C)	For the equilibrium $H_2O(g)+CO(g) \rightleftharpoons H_2(g)+CO_2(g)$, if inert gas is added at constant pressure at equilibrium	(r)	Backward shift
(D)	For the equilibrium $PCl_5 \rightleftharpoons PCl_3 + Cl_2$ if more PCl_5 is added		

(1) A – (p), B – (q), C – (r), D – (q)

(2) A – (r), B – (r), C – (q), D – (p)

(3) A – (p), B – (p), C – (q), D – (r)

(4) A – (q), B – (r), C – (r), D – (p)

37) Suggest the way by which the concentration of hydrazine (N_2H_4) could be increased in an equilibrium described by the equation :-
$N_2(g) + 2H_2(g) \rightleftharpoons N_2H_4(g)$; $\Delta H = 95$ kJ

(1) Increasing temperature

(2) Increasing pressure

(3) Decreasing volume

(4) All of the above

38) For a reaction; $A + B \rightleftharpoons C + D$, the initial concentration of A and B are equal but the equilibrium concentration of C is twice that of equilibrium concentration of A. Then K_C is :

(1) 4

(2) 9

(3) 1/4

(4) 1/9

39) Equal moles of H_2 and I_2 are heated in a closed vessel till equilibrium is obtained. If 80% of the hydrogen is converted to HI, the K_c at this temperature is ($H_2 + I_2 \rightleftharpoons 2HI$)

(1) 64

(2) 16

(3) 0.25

(4) 14

40) For the given reaction,
$A(s) \rightleftharpoons 2B_{(g)} + C_{(g)}$
If at equilibrium concentration of B is doubled, then the concentration of C at equilibrium will be :

(1) Doubled

(2) Halved

(3) One-forth of original concentration

(4) One-eight of original concentration

41) A mixture of A, B and C gases is maintained in a 10 litre flask at a temperature at which the equilibrium constant K_c of the reaction :
$2B(g) + C(g) \rightleftharpoons 2A(g)$ is 100
If the number of moles of B and A in the flask are equal at equilibrium, then number of moles of C present in flask is :-

(1) 0.01

(2) 0.1

(3) 1.0

(4) 10.0

42) At 1000 K, the value of K_P for the reaction
$PCl_5 \rightleftharpoons PCl_3 + Cl_2$ is 20R, then K_C will be :-

(1) 0.04

(2) 0.02

(3) 0.01

(4) 0.03

43) K_C for $PCl_5 \rightleftharpoons PCl_3 + Cl_2$, is 0.04 at 25°C. How many moles of PCl_5 must be added to a 3 litre flask to obtain a Cl_2 concentration of 0.15 M :-

(1) 1.2

(2) 2.1

(3) 1.8

(4) 0.7

44)

When heated, ammonium carbamate decomposes as follows :
$NH_4COONH_2(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$
At a certain temperature, the equilibrium pressure of the system is 0.318 atm. K_p for the reaction is:

(1) 0.128

(2) 0.426

(3) 4.76×10^{-3}

(4) 4.26×10^{-2}

45) Active mass of 10 gm $MgCO_3$ (s) is :-

(1) 1

(2) $\frac{10}{84}$

(3) 10

(4) data insufficient

BIOLOGY

1)

Consider the following four statements (A-D) and select the option which includes all the **correct** ones only :-
A.The portion of the embryonal axis above the level of cotyledons is the hypocotyl which terminates with the plumule
B. Seeds of beet possesses remnant of nucellus called perisperm.
C. Parthenocarpy can be induced through the applications of growth hormones.
D. Seeds have better adaptive strategies for dispersal to new habitates and help the species to colonise in other areas.
Options :-

(1) Statements A, B, C

(2) Statements B, C, D

(3) Statements A, B, D

(4) Statements A, B, C, D

2)

In a transverse section, a typical microsporangium appears near _____ in outline :-

(1) Tetragonal

(2) Circular

(3) Polygonal

(4) Tetrahedral

3) In angiosperms, the following events occur during megasporogenesis and megagametogenesis –
(i) Meiosis in megaspore mother cell
(ii) Formation of a linear tetrad
(iii) Fusion of the polar nuclei
(iv) Formation of the embryo sac
(v) Differentiation of the megaspore mother cell
(vi) Degeneration of 3-megaspores
(vii) Mitotic divisions in functional megaspore
The correct sequence of events is :-

(1) (i), (ii), (iii), (iv), (v), (vi), (vii)

(2) (v), (i), (ii), (vi), (iv), (iii), (vii)

(3) (v), (i), (ii), (vii), (vi), (iii), (iv)

(4) (v), (i), (ii), (vi), (vii), (iv), (iii)

4) During an experiment, vegetative nucleus was destroyed inside the pollen tube. Immediately afterwards it was observed that the growth of pollen tube was stopped and pollen tube could not reach up to ovule, because-

(1) Male gametes were not developed

(2) Growth of pollen tube is controlled by vegetative nucleus

(3) Generative nucleus secretes chemicals

(4) All of the above

5) The number of chromosomes in radicle is 16. What will be the number of chromosomes in tube nucleus, antipodal cells, secondary nucleus and endosperm respectively ?

(1) 8, 8, 16, 24

(2) 8, 8, 16, 16

(3) 16, 16, 32, 48

(4) 8, 8, 24, 48

6) (A) Polar nuclei are situated below the egg apparatus in the large central cell
(B) Filiform apparatus play important role in guiding the entry of pollen tube into embryosac

(1) Both (A) and (B) are correct

(2) Both (A) and (B) are incorrect

(3) Only (A) is correct

(4) Only (B) is correct

7) Which statement is incorrect regarding *Parthenium*?

(1) It came into India as contaminant with imported wheat.

(2) It's pollen cause allergy.

(3) Now it becomes ubiquitous in occurrence.

(4) It is also known as sea grass.

8) Match a, b, c and d with i, ii, iii and iv:

(a)	Pollen tablets	(i)	Pollen viability for months
(b)	Rosaceae	(ii)	Pollen Allergy
(c)	Pollen bank	(iii)	Food supplement
(d)	*Parthenium*	(iv)	–196°C

Choose the correct match :

(1) a-i, b-iii, c-ii, d-iv

(2) a-iii, b-iv, c-i, d-ii

(3) a-iii, b-i, c-iv, d-ii

(4) a-ii, b-i, c-iv, d-iii

9) Which of the following is not correct?

(1) Microspore mother cell $\xrightarrow{\text{Meiosis}}$ Microspores

(2) Microspore $\xrightarrow{\text{Unequal Mitosis}}$ mature pollen grain

(3) Megaspore mother cell $\xrightarrow{\text{Meiosis}}$ Megaspores

(4) Megaspore $\xrightarrow{\text{Meiosis}}$ Female gametophyte

10) Choose the statements which are not correct for angiosperm :-
(a) Male gametes are motile.
(b) Pollen tube always enters into embryo-sac through the degenerating antipodal.
(c) Double fertilization is specific character of dicots only.

(1) a, c

(2) b, c

(3) a, b, c

(4) a, b, c

11) Which of the following statement is not correct :

(1) Formation of fruit from unfertilized ovary is known as parthenocarpy

(2) Female gamete of angiosperms is non-motile

(3) Generative cell floats in the cytoplasm of vegetative cell

(4) Stamens are attached to petals or tepals from distal end of filament

12) Which of the following statement is not correct for pollination ?

(1) In monoecious plant, autogamy is prevented but not geitonogamy

(2) In cleistogamous flower autogamy and geitonogamy both can happen

(3) *Hydrilla* shows hypohydrophily

(4) Wind pollination is quite common in grasses.

13) Find the correct group regarding endospermic or albuminous seeds :–

(1) Castor, maize, wheat

(2) Pea, castor, orchids

(3) Maize, gram, pea

(4) Pea, barley, gram

14) If the number of chromosomes in the leaf mesophyll cells of maize plant is 20. What would be the number of chromosomes in its endosperm, generative cell and nucellus respectively ?

(1) 10, 20, 30

(2) 30, 10, 20

(3) 20, 30, 10

(4) 30, 20, 20

15) Which of the following statement is not correct about geitonogamy ?

(1) Pollination takes place in between the two flowers of the same plant.

(2) Genetically, it is self pollination
(3) Functionally, it is cross pollination
(4) Genetically as well as functionally it is cross pollination

16) Microsporangium is generally surrounded by how many wall layers :-

(1) One
(2) Two
(3) Three
(4) Four

17) A typical female gametophyte of angiosperm consists of :-

(1) Egg cell, synergids and antipodal cells only.
(2) Egg apparatus, antipodal cells and central cell only.
(3) Egg apparatus and antipodal cells only.
(4) Antipodal cells and synergid cell only.

18) Non- albuminous seeds are found in :-

(1) Wheat
(2) Maize
(3) Pea
(4) Barley

19) Yucca plant is pollinated by which agent :-

(1) Wasp
(2) Moth
(3) Bees
(4) Water

20) Both male and female flowers are present on the different plants in :-

(1) Maize and papaya
(2) Castor and papaya
(3) Maize and castor
(4) Papaya and date palm

21) The body of the ovule fuses with funicle in the region called :-

(1) Caruncle
(2) Hilum
(3) Raphe
(4) Obturator

22) As the anthers mature and dehydrate the microspores dissociate from each other and develop into :-

(1) Megaspores
(2) Seed
(3) Pollen grains
(4) Embryo sac

23) Testa of a seed is derived from :-

(1) Ovary wall
(2) Hilum
(3) Nucellus
(4) Outer integument of ovule

24) The coconut water of tender coconut is :-

(1) Cellular endosperm
(2) Free nuclear endosperm
(3) Helobial endosperm
(4) Polygonum endosperm

25) In angiosperms, triple fusion is required for the formation of :-

(1) Fruit wall
(2) Seed coat
(3) Embryo sac
(4) Endosperm

26) Filiform apparatus is present in :

(1) Synergids
(2) Pollen grain
(3) Nucellus
(4) Endosperm

27) Majority of the flowering plants use :-

(1) Biotic agents for pollination
(2) Abiotic agents for pollination
(3) Wind for pollination
(4) Water for pollination

28) What is the end product of sexual reproduction ?

(1) Ovule

(2) Androecium
(3) Seed
(4) Ovary

29) In angiosperms megasporangium is commonly called as :-

(1) Ovule
(2) Embryo sac
(3) Carpel
(4) Gynoecium

30) Identify the characters unique to angiosperm :-

(1) Presence of xylem
(2) Presence of ovule
(3) Presence of vascular bundles
(4) Double fertilization

31) Seed formation without fertilization in flowering plants is called :-

(1) Budding
(2) Fragmentation
(3) Amphimixis
(4) Apomixis

32) Excavated oldest viable seed is of :-

(1) *Phoenix dactylifera*
(2) *Lupinus arcticus*
(3) *Pisum sativum*
(4) *Papaver somniferum*

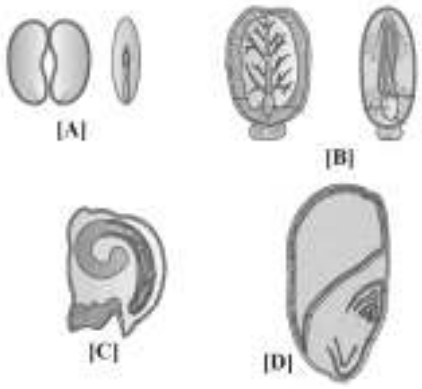
33) In which of the following plant many embryos are developed from nucellus :-

(1) Mango
(2) Citrus
(3) Both (1) and (2)
(4) Wheat

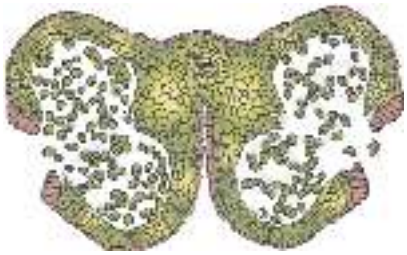
34) **Statement-I :** In *Vallisneria* pollen grains are carried to the stigma by water current.
Statement-II : *Vallisneria* is a dioecious plant.

(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

35) Which one of the following diagrams represents endospermic dicot seed?

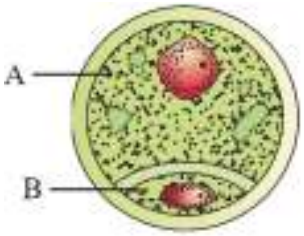


(1) A only
(2) B only
(3) C, D both
(4) A, B both



36) How many wall layers are found in a pollen sac at following stage.

(1) 2
(2) 3
(3) 4
(4) 1



37) What are the parts A and B in this figure ?

(1) A : Generative cell, B : Vegetative cell
(2) A : Tapetum cell, B : Generative cell
(3) A : Vegetative cell, B : Generative cell
(4) A : Vegetative cell, B : Tube cell

38)

The given plant is pollinated by :-



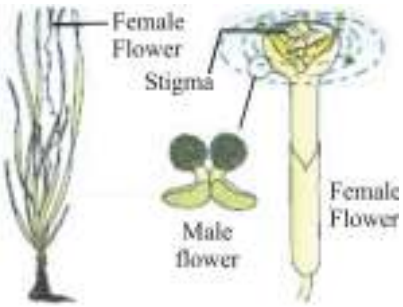
(1) Wind

(2) Water

(3) Insect

(4) Bird

39) Given below is the diagram of pollination in plants. Which type of pollination is represented by this figure ?



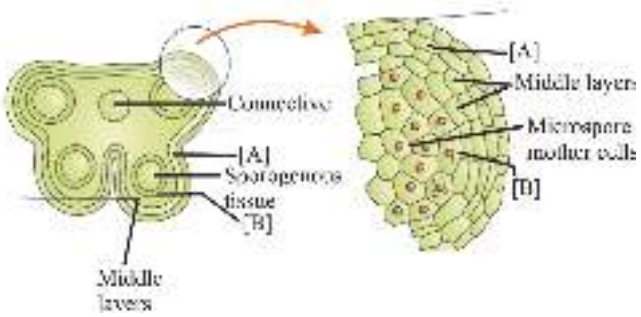
(1) Anemophily

(2) Entomophily

(3) Hydrophily

(4) Ornithophily

40) Transverse section of a young anther is given below. Identify labelled structures [A] & [B] :-



(1) [A] - Epidermis; [B] - Endothecium

(2) [A] - Endothecium; [B] - Epidermis

(3) [A] - Endothecium; [B] - Tapetum

(4) [A] - Tapetum; [B] - Endothecium

41) **Assertion(A) :-** Seeds of groundnut are ex-albuminous.
Reason (R) :- Endosperm is completely consumed during embryonic development.

Select the correct option :

(1) Assertion is true but Reason is Wrong

(2) Assertions wrong and Reason is true

(3) Both Assertion and Reason are correct and reason is correct explaination for Assertion

(4) Both Assertion and reason are correct but Reason is not correct explaination of Assertion

42) **Assertion:** The ability of the pistil to recognise the pollen is the result of a continuous dialogue between pollen grain & pistil.
Reason: The dialogue is mediated by physical components of pollen interacting with those of pistil.

(1) Both Assertion & Reason are correct & Reason is correct explanation of Assertion.

(2) Both Assertion & Reason are correct but Reason is not correct explanation of Assertion.

(3) Only Assertion is correct but Reason is incorrect.

(4) Both Assertion & Reason are incorrect.

43) **Statement - I :-** Pollen grains of many species cause severe allergies and bronchial afflictions in some people.
Statement - II :- Pollen grains are rich in nutrients and they are used in the form of tablets and syrups.

(1) Both statements - I & II are incorrect.

(2) Both statements - I & II are correct.

(3) Statements - I is correct & II is incorrect.

(4) Statement - I is incorrect & II is correct.

44) Given below are two statements :
Statement-I :- All flowering plants show sexual reproduction.
Statement-II :- Flowers are sites of sexual reproduction.
Choose the correct answer from the options given below :-

(1) Both statement-I & II are correct.

(2) Both statement-I & II are incorrect.

(3) Statement-I is incorrect but statement-II is correct.

(4) Statement-I is correct but statement-II is incorrect.

45) Given below are two statements :
Statement-I : A typical angiosperm anther is bilobed with each lobe having two theca.
Statement-II : The innermost layer of anther wall is known as tapetum.
In the light of above statements choose the correct answer from the options given below :

(1) Both statement I & II are true.

(2) Statement I is true & Statement II is false.

(3) Statement I is false & Statement II is true.

(4) Both Statement I & II are false.

46)

Consider the following statements (A–D) and select the option which includes all the **correct** ones :-

(A) A leaf is said to be simple when its lamina is entire or when incised, the incision do not touch the mid rib or petiole. (B) A compound leaf has axillary bud in its axil.
(C) Alternate phyllotaxy is found in *Calotropis* and Guava. (D) Reticulate venation is found in mango.

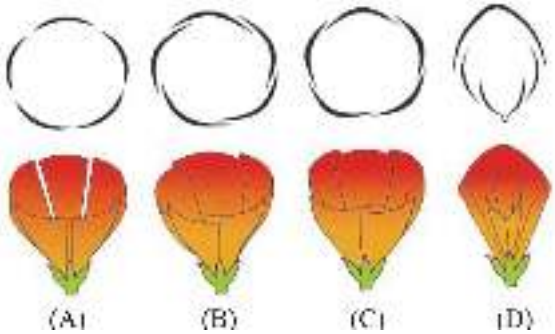
(1) A, B, C

(2) A, B

(3) A, B, D

(4) A and C

47) The aestivation in petals of lady finger and *Calotropis* plants are respectively :-



(1) A & D

(2) B & A

(3) D & A

(4) D & B

48)

Consider the following 4 statements a-d and select the option which includes all the **correct** ones :-
(a) When stamens are united by their filaments only then it is called adelphy.
(b) When anthers are united and filaments are free, then it is syngenesious condition.
(c) When anthers as well as filaments are united then it is called synandrous condition.

(d) In tetradynamous condition inner 4 stamens are short while outer 2 stamens are long.

(1) Statement a, b, c

(2) Statement b, c, d

(3) Statement b, d

(4) Statement c, d

49) Dicot plants differ from monocots in many characters like :-

(1) In dicots two cotyledons are found in seeds while in monocots plants, there are either 3 or multiple of three.

(2) Sheathing leaf base is found in dicot plants while pulvinus leaf base in monocots plants

(3) Parallel venation is found in dicots while reticulate in monocot.

(4) In dicots, flowers are tetra or pentamerous while trimerous in monocots.

50) Study the sentences carefully and give the answer.
(A) When a flower can be divided into two equal radial halves through any plane, it is zygomorphic.

(B) Mustard, datura and chilli are examples of actinomorphic flowers
(C) Pea, gulmohur, bean, *Cassia* are examples of zygomorphic flowers. (D) If a fruit is formed without fertilisation of the ovary, it is called geocarpic fruit.

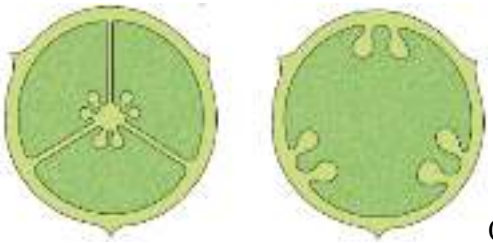
(1) Both A and B are correct
(2) Both C and D are correct
(3) Both A and D are incorrect
(4) Both B and D are incorrect

51) Select the mismatched pair from the following :-

(1) Whorled phyllotaxy – *Alstonia*
(2) Pulvinus leaf base – Wheat
(3) Palmately compound leaf – Silk cotton
(4) Sheathing leaf base – Banana

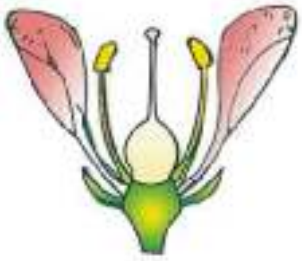
52) In some_____the leafbase may become swollen and is called_____.

(1) Monocot, sheathing leaf base
(2) Legumes, pulvinus leaf base
(3) Legumes, sheathing leafbase
(4) Monocots, pulvinus leaf base

53)  Given diagrams represent :-

(1) Marginal and parietal placentation respectively
(2) Axile & parietal placentation respectively
(3) Parietal & axile placentation respectively
(4) Free central & parietal placentation respectively

54) According to given diagram, find out the type of flower and position of ovary :-



(1) Epigynous - ovary superior
(2) Hypogynous - ovary inferior
(3) Hypogynous - ovary superior

(4) Epigynous - ovary inferior

55) Which of the following is/are not common for both coconut and mango ?

(1) Presence of stony endocarp
(2) Single seeded fruit
(3) Thick meoscarp
(4) Edible Endocarp

56) Which one of these is not a character of solanaceae family ?

(1) Swollen placenta and oblique septum
(2) Many non endospermic seeds
(3) Epipetalous stamens
(4) Persistent calyx

57) Which of the following is an **incorrect** match ?

(1) Perigynous flower – Pulm, rose and peach
(2) Monoadelphous stamens – Pea
(3) Epigynous flower – Guava, cucumber and ray florets of sunflower
(4) Polyadelphous stamens – *Citrus*

58) Superior ovary is found in :-

(1) Compositae and cruciferae
(2) Cucurbitaceae and malvaceae
(3) Cucurbitaceae and fabaceae
(4) Malvaceae and liliaceae

59) Study the following statements carefully and give the answer :-
(A) Petioles allow leaf blades to flutter in wind
(B) Veins provide rigidity to the leaf blade
(C) A bud is present in the axil of both leaf and leaflet
(D) In pinnately compound leaf, all leaflets are attached to a common point i.e. tip of petiole

(1) Both A and C are correct
(2) Both A and D are correct
(3) Both C, D are wrong
(4) Both B and C are wrong

60) $\oplus \male\!\female\ K_{(5)}\,\overset{\frown}{C_{(5)}}\,A_5\,\underline{G}_{(2)}$
is the floral formula of which plant ?

(1) Mustard
(2) Onion

(3) Potato

(4) Pea

61) **Assertion:** The cymose type of inflorescence has limited growth.
Reason: In cymose inflorescence the main axis terminates in a flower.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

62) **Assertion:** Gulmohar shows bilateral symmetry of flowers.
Reason: Gulmohar flowers can be divided into two similar halves only in one particular vertical plane.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

63) **Assertion:** Potato tuber is a modified stem.
Reason: Potato tuber contains nodes and internodes.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

64) **Assertion :** Reticulate venation is found in dicots
Reason : Veinlets are absent in reticulate venation

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

65)  Above type of phyllotaxy can be seen in :-

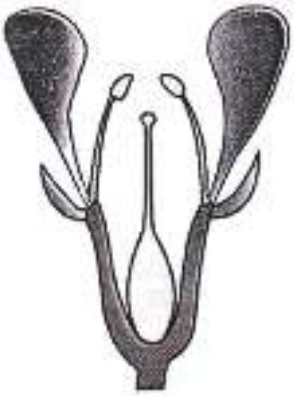
(1) Guava

(2) Chinarose

(3) *Alstonia*

(4) *Nerium*

66) Identify the given diagram and select the correct option :-



(1) Epigynous flower
(2) Half inferior ovary
(3) Perigynous flower
(4) Both (2) and (3)

67) Which one of the following diagram represents the placentation in tomato :-

(1) 

(2) 

(3) 

(4) 



68) Following type of aestivation in petals can be seen in which plant :-

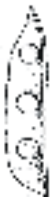
(1) *Calotropis*
(2) Cotton
(3) *Cassia*
(4) Mustard

69) Which one of the following diagrams represents the placentation in *Dianthus* ?

(1) 

(2) 

(3) 

(4) 

70) Bicarpellary gynoecium with ovaries arranged obliquely on thalamus occurs in :

(1) *Capsella*
(2) Brinjal
(3) China rose
(4) Pea

71) When the anthers are united but filaments remain free, Such an androecium is known as :

(1) Synandrous
(2) Polyadelphous
(3) Syngenesious
(4) Monoadelphous

72) Among china rose, mustard, rose, brinjal, guava, plum, ray florets of sunflowers, cucumber; How many plants have completely inferior ovary?

(1) Five
(2) Four
(3) Six
(4) Three

73) Which plants contain variations in length of filaments of androecium within their flower ?

(1) *Citrus* and *Salvia*
(2) Mustard and china rose
(3) China rose and *Salvia*
(4) Mustard and *Salvia*

74) In a flower, if a whorl member overlaps without any direction, than aestivation is known as :

(1) Vexillary
(2) Imbricate
(3) Twisted
(4) Valvate

75) When an ovary develops into fruit without fertilization, this phenomenon is known as :

(1) Autogamy
(2) Parthenocarpy
(3) Syngamy
(4) Parthenogenesis

76) The flowers are actinomorphic in :

(1) Gulmohar
(2) *Datura*
(3) Bean
(4) *Cassia*

77) When the placenta is axial and the ovules are attached to it in a multilocular ovary, the placentation is said to be :

(1) Basal
(2) Marginal
(3) Parietal
(4) Axile

78) When margins of petals just touch to each other but do not overlap then it is known as .....A..... aestivation and it is present in .....B..... . A and B are :

(1) A : Twisted ; B : China rose
(2) A : Valvate ; B : *Calotropis*
(3) A : Valvate ; B : China rose
(4) A : Imbricate ; B : *Cassia*

79) In *Solanum nigrum* the flowers are :-

(1) Zygomorphic, epigynous and pentamerous
(2) Actinomorphic, epigynous and pentamerous
(3) Actinomorphic, hypogynous and pentamerous
(4) Zygomorphic, hypogynous and tetramerous

80) Mango has

(1) Undifferentiated pericarp
(2) Fibrous and thin mesocarp

(3) Stony endocarp
(4) Edible endosperm

81) Epiphyllous stamens are seen in

(1) Soyabean
(2) Lily
(3) Mustard
(4) Brinjal

82) Mango and coconut are similar to each other in all, except

(1) Both have edible mesocarp
(2) Both develop from monocarpellary ovaries.
(3) Both develop from superior ovaries
(4) Both have hard endocarp.

83) Select the odd one w.r.t. epigynous flowers

(1) Guava
(2) Rose
(3) Cucumber
(4) Ray florets of sunflower

84) When stamens are attached to the perianth then it is called

(1) Monadelphous
(2) Epipetalous
(3) Diadelphous
(4) Epiphyllous

85) Match the column-I and column-II and choose the correct option :-

	Column-I		**Column-II**
(i)	Pneumatophores	(a)	*Citrus*
(ii)	Stem tendrils	(b)	Legumes
(iii)	Stem thorns	(c)	*Rhizophora*
(iv)	Pulvinus	(d)	Grape vine

(1) (i)-(a), (ii)-(b), (iii)-(d), (iv)-(c)
(2) (i)-(c), (ii)-(a), (iii)-(d), (iv)-(b)
(3) (i)-(d), (ii)-(c), (iii)-(a), (iv)-(b)
(4) (i)-(c), (ii)-(d), (iii)-(a), (iv)-(b)

86) Belladona and ashwagandha belong to :-

(1) Brassicaceae
(2) Fabaceae
(3) Solanaceae
(4) Liliaceae

87) Apocarpous ovary is found in :-

(1) Lotus and rose
(2) Mustard and tomato
(3) Radish and *capsella*
(4) Tomato and potato

88) Which of the following is not related to family solanaceae ?

(1) Parietal placentation
(2) Swollen placenta with many ovules
(3) Alternate phyllotaxy
(4) Epipetalous stamens

89) When petals in a whorl just touch one other at the margin without overlapping. This type of aestivation is present in :-

(1) *Calotropis*
(2) *Cassia*
(3) Gulmohur
(4) Bean

90) Pappus in sunflower family is the modification of

(1) Hairs
(2) Anthers
(3) Calyx
(4) Corolla

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	4	2	3	3	3	1	3	3	2	1	4	1	2	1	1	3	1	2	1	4
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	1	3	2	3	2	4	1	4	3	4	2	2	2	3	2	2	4	3	4
Q.	41	42	43	44	45															
A.	4	1	1	4	4															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	1	2	1	3	1	2	3	3	1	3	1	2	1	3	3	1	2	3	2	3
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	3	3	3	2	2	1	4	2	2	1	2	3	2	1	1	2	4	1	1	3
Q.	86	87	88	89	90															
A.	2	2	2	3	1															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	2	2	4	2	1	1	4	3	4	3	4	2	1	2	4	4	2	3	2	4
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	2	3	4	2	4	1	1	3	1	4	4	2	3	1	2	1	3	1	3	3
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	3	2	1	1	3	2	1	4	3	2	2	2	3	4	2	2	4	3	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	1	1	1	3	1	4	3	3	3	2	3	4	4	2	2	2	4	2	3	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	2	1	2	4	4	3	1	1	1	3										

SOLUTIONS

PHYSICS

1) $\frac{dr}{dt} = 0.1$ $A = \pi r^2$

$$\frac{dA}{dt} = 2\pi r \cdot \frac{dr}{dt} = \frac{2\pi \times 5}{11} \times 0.1$$

$= 0.29\ cm^2/sec$

2) $\sin 31° = \sin (30° + 1°)$

$\because \sin (A + B) = \sin A \sin B + \cos A \cos B$

$\Rightarrow \sin (30° + 1°) = \sin 30° \cos 1° + \sin 1° \cos 30°$

$\cos 1° \approx 1$, $\sin 1° = \sin\left(\frac{\pi}{180}\right) = \frac{\pi}{180}$ (small angle)

$$= \frac{1}{2} \times 1 + \frac{\pi}{180} \times \frac{\sqrt{3}}{2} = \frac{1}{2}\left(1 + \frac{\pi\sqrt{3}}{180}\right)$$

3) $P^2 + Q^2 = 2(F_1^2 + F_2^2)$

$P^2 + Q^2 = 100$

$\sqrt{P^2 + Q^2} = 10$

4) $\left(2\hat{i} + 2\hat{j} - 2\hat{k}\right) \times \left(-\hat{i} + 2\hat{j} + 2\hat{k}\right) = \vec{A}$ (say)

$\Rightarrow \vec{A} = 8\hat{i} - 2\hat{j} + 6\hat{k}$

Let, $5\hat{i} + y\hat{j} + \hat{k} = \vec{B}$ (say)

Since given three vector are coplanar $\vec{A}$ and $\vec{B}$ are $\perp$ to each other.

Therefore, $\vec{A}.\vec{B} = 0$

$\Rightarrow 40 - 2y + 6 = 0 \Rightarrow y = 23$

5)

Angle made by $\vec{A}$ with x-axis

$$\cos\alpha = \frac{A_x}{A}$$

$$\cos\alpha = \frac{4}{\sqrt{4^2 + 3^2 + 12^2}} = \frac{4}{13}$$

6) $\vec{F}_x = 1\cos 60° \hat{i} - 4 \sin 30° \hat{i} + 2 \sin 30° \hat{i}$

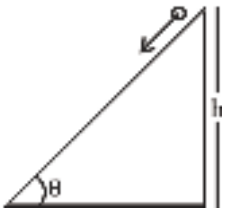
$$= \frac{1}{2}\hat{i} - 2\hat{i} + \hat{i}$$

So minimum additional force required is

$\vec{F} = -\vec{F}_x = 0.5\hat{i}$

7) $\frac{h}{\sin\theta} = \frac{1}{2}g\sin\theta t^2$

$t = \left(\sqrt{\frac{2h}{g}}\right)\frac{1}{\sin\theta}$



$\theta_2 > \theta_1 \Rightarrow t_2 < t_1$

8)

As $v^2 = u^2 - 2as \Rightarrow u^2 = 2as$ $(\because v = 0)$

$\Rightarrow \quad u^2 \propto s \Rightarrow \frac{u_2}{u_1} = \left(\frac{s_2}{s_1}\right)^{1/2}$

$\Rightarrow \quad u^2 = \left(\frac{9}{4}\right)^{1/2} u_1 = \frac{3}{2}u_1 = 300 \text{ cm/s}$

9)

let the time interval between two drops is 't'

$8t = \sqrt{\frac{2 \times 16}{10}}, \quad t = \sqrt{\frac{1}{20}}$

distance between 3rd & 5th drop

t ↕ 0 9th
0 8th

t ↕ 0 7th

t ↕ 0 6th

t ↕ 0 5th

t ↕ 0 4th

t ↕ 0 3rd

t ↕ 0 2nd

t ↕ 1st

$= \frac{1}{2}g(6t)^2 - \frac{1}{2}g(4t)^2$

$= \frac{1}{2}g \times t^2(36 - 16) = \frac{1}{2} \times 10 \times \frac{1}{20} \times 20$

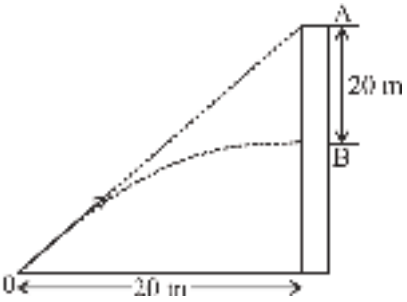
$= 5\text{m}$

10) $x = 4 - 2t + t^2$

$v = -2 + 2t = 2t - 2$

Speed $= |2t - 2|$

11) $\frac{d\vec{v}}{dt} = \vec{a} = \vec{g} = \text{constant}$

12) 

Distance between A and B = 20 m

$20 = \frac{1}{2} g t^2$

$t = 2$ sec

$V_x = \frac{S_x}{t} = \frac{20}{2}$

$= 10$ m/sec

13) $S_{rel} = u_{rel} t + \frac{1}{2} a_{rel} t^2$

$100 + 125 = (10 + 10)\, t + \frac{1}{2} (0.3 + 0.2)\, t^2$

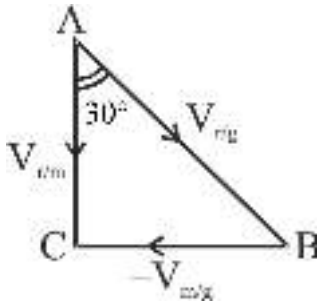
$900 = 80\, t + t^2$

$\Rightarrow \quad t^2 + 90\, t - 10\, t - 900 = 0$

$t = -90,\ t = 10$ sec

14) $|\vec{V}_{AB}| = \sqrt{30^2 + 40^2} = 50$

$S_{AB} = 50 \times 2 = 100$ m

15) 

$\sin 30^\circ = \frac{V_{m/g}}{V_{r/g}}$

$V_{r/g} = 2V_{m/g} = 20$ km/hr

16)

$$\therefore t = \frac{d}{\sqrt{v_b^2 - v_R^2}} = \frac{2}{\sqrt{(20)^2 - (10)^2}} = \frac{2}{10\sqrt{3}}$$

$$\therefore t = \frac{1}{5\sqrt{3}} h$$

18) Reading = M.S.R + L.C. × V.S.R

[L.C. = $\left(\frac{20-19}{20}\right) 0.20$ cm = 0.01 cm]

$= (5.1cm) + (6 \times 0.01cm)$

= 5.16 cm

19)

Error in AB

$= \pm \left(\frac{\Delta A}{A} + \frac{\Delta B}{B}\right) AB = \pm \left[\frac{0.5}{2.5} + \frac{0.01}{0.10}\right] [0.25]$

= ± 0.075 = ± 0.08

Therefore, AB = (0.25 ± 0.08) m

20) Dipole moment = charge × distance

Electric flux = electric field × area

∴ Dipole moment, electric flux and electric field have different dimensions.

21) Logarithm has no dimensions.

$[xb] = [M^0L^0T^0] \Rightarrow [b] = [L^{-1}]$

Hence, option (2) is correct.

22) $\nu \propto \sigma^a \delta^b \lambda^c$

$[M^\circ L\ T^{-1}] = [M\ T^{-2}]^a\ [ML^{-3}]^b\ [L]^c$

$-1 = -2a \Rightarrow a = \frac{1}{2}$

$o = a + b \Rightarrow b = -\frac{1}{2}$

$1 = -3 \times \left(\frac{-1}{2}\right) + c \Rightarrow c = \frac{-1}{2}$

$\nu \propto \sqrt{\frac{\sigma}{\delta\lambda}} \Rightarrow \nu^2 \propto \frac{\sigma}{\delta\lambda}$

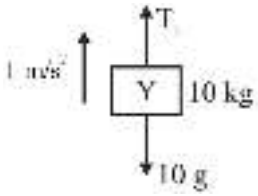
23) a + b = 1.00

a – b = 0.84

In multiplication and division result will contain least no. of significant figure.

24) $T_1 - 10g = 10(1)$

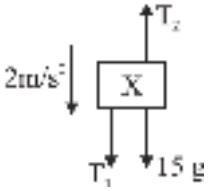
$T_1 = 110$



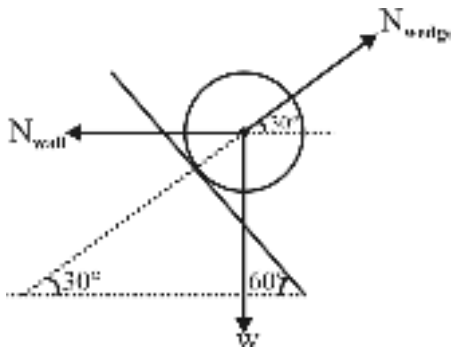
$T_1 + 15g - T_2 = 15(2)$

$110 + 150 - T_2 = 30$

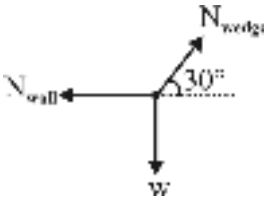
$T_2 = 230$ N



25)



OR



$N_{wedge} \sin 30° = w$

$N_{wall} = (N_{wedge}) \cos 30°$

$= (2w)\frac{\sqrt{3}}{2} = w\sqrt{3} = 5g\sqrt{3}$

$N_{wall} = 50\sqrt{3} = F$

26) $a = \frac{60 - 18 - (3+2+1)g\sin 30°}{(3+2+1)}$

$a = \frac{12}{6} = 2$ m/s^2 upwards.

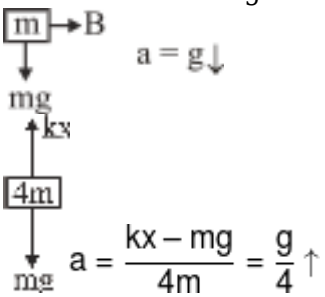


27) $T = mg$



kx = T + 4 mg = 5mg

After cut the string T = 0



28) Component of $\frac{10}{\sqrt{3}}$ along the inclined plane is just sufficient to balance component of weight along the inclined plane. So there will be no tendency of slipping. Therefore. Friction = 0.

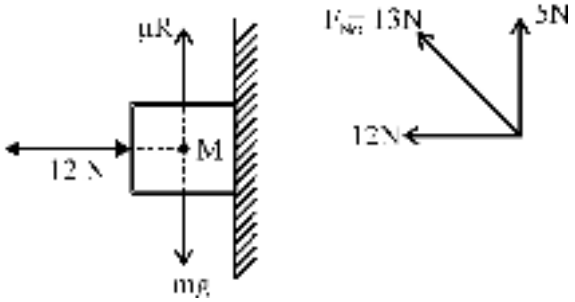
29) N = F = 12 N

Static frictional force = μN = 0.5 × 12 = 6 N

mg = 5N

so friction = F_S = 5N

$F_{Net} = \sqrt{5^2 + 12^2}$ = 13 N



30)

Frictional force on block B

= $\mu m_B g$ = 0.4 × 3 × 10 = 12 N

Hence, maximum common acceleration = $\frac{12}{3} = 4ms^{-2}$

Hence, maximum force

F = $(m_A + m_B)a$ = (6 + 3) × 4 = 36N

31) $m_1 = -\left(\frac{f}{u_1 - f}\right)$

$m_2 = -\left(\frac{f}{u_2 - f}\right)$

$$\frac{m_1}{m_2} = \frac{u_2 - f}{u_1 - f}$$

32) $$\frac{L}{t} = \frac{\sin(i-r)}{\cos r} = \frac{1}{\sqrt{3}}$$

$$\frac{\sin 45 \cos r}{\cos r} - \frac{\cos 45 \sin r}{\cos r} = \frac{1}{\sqrt{3}}$$

$$1 - \tan r = \sqrt{\frac{2}{3}}$$

$$r = \tan^{-1}\left(1 - \sqrt{\frac{2}{3}}\right)$$

33) $\sqrt{2}\sin r_2 = \sin 90^\circ$

$r_2 = 45^\circ$

$r_1 = 30^\circ$

$\sin i = \sqrt{2}\sin 30^\circ$

$i = 45^\circ$

34) $$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

Here $\mu_2 = \mu$, $\mu_1 = 1$

For 1st focus $u = f_1$

$v = \infty$ $\Rightarrow \frac{\mu}{\infty} - \frac{1}{f_1} = \frac{\mu - 1}{R} \Rightarrow f_1 = \frac{-R}{\mu - 1}$

For 2nd focus $u = \infty$

$v = f_2$ $\Rightarrow \frac{\mu}{f_2} - \frac{1}{\infty} = \frac{\mu - 1}{R} \Rightarrow f_2 = \frac{\mu R}{\mu - 1} \Rightarrow f_2 + \mu f_1 = 0$

35) $$\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\frac{1}{f} \propto \mu \propto \frac{1}{\lambda}$$

Assertion is true reason is false

36) For an object placed at infinity the image after first refraction will be formed at v_1

$$\frac{\mu_2}{v_1} - \frac{\mu_1}{-\infty} = \frac{\mu_2 - \mu_1}{+R} \quad \text{... (i)}$$

The image after second refraction will be found at v_2

$$\frac{\mu_3}{v_2} - \frac{\mu_2}{v_1} = \frac{\mu_3 - \mu_2}{+R} \quad \text{... (ii)} \quad \text{adding (i) and (ii)}$$

$$\frac{\mu_3}{v_2} = \frac{\mu_3 - \mu_1}{R} \Rightarrow v_2 = \frac{\mu_3 R}{\mu_3 - \mu_1}$$

Now, for lens $\frac{1}{v_2} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v_2} - \frac{1}{-\infty} = \frac{1}{f} \Rightarrow v_2 = f$

Therefore focal length will be $\frac{\mu_3 R}{\mu_3 - \mu_1}$ = 2R

37) $m = \frac{f}{u+f}$

$m' = \frac{f}{-x+f}$, $m = \frac{f}{-y+f}$

⇒ m' = –m

⇒ $\frac{f}{-y+f} = \frac{-f}{-x+f}$ □ y -f = f -x

⇒ $f = \left(\frac{x+y}{2}\right)$

38)

f_0 = 20 m = 2000 cm

f_e = 2 cm

for Normal adjustment

→ M.P. = $-\frac{f_0}{f_e} = \frac{-2000}{2} = -1000$

→ Distance between both lens = $f_0 + f_e$

= 2000 + 2

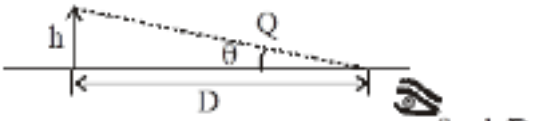
= 2002 cm

= 20.02 m

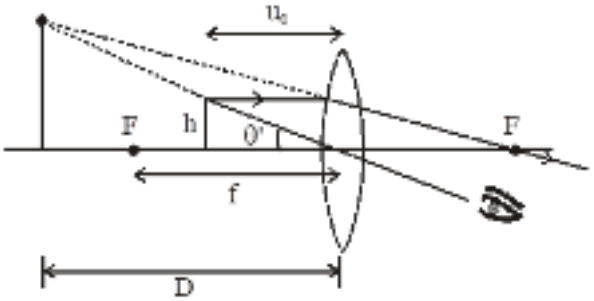
→ Image is inverted and magnified

→ Aperture of eye piece is smaller than objective.

39)



θ = h/D



$\theta' = \frac{h}{u_0}$; θ' is same for both object and image

$m = \frac{\theta'}{\theta} = \frac{D}{\mu_0}$

u_0 < D

Hence m > 1

40) When the lens of simple microscope is placed near an object, one focal length away or less and the eye is positioned close to the lens on the other side. Then, an erect, magnified and virtual image of the object can be seen comfortably, i.e. at 25 cm or more. Also, the angle substended by the object is the same as that at the near point.
If the object is at a distance f, the image is at infinity. However, if the object is at a distance sightly less than the focal length of the lens, the image is virtual and closer than infinity.
Hence, A → 2, B → 3 and C →1.

41)

d = 1mm, λ = 6000 Å , D = 1 m
$I_{max} = 4\ I_0$
⇒ 75% of Imax = $3\ I_0$
$3I_0 = I_0 + I_0 + 2\sqrt{I_0 I_0}\cos(\Delta\theta)$
$I_0 = 2I_0\cos\Delta\theta$
$\cos\Delta\theta = \frac{1}{2}$
$\Delta\theta = \pm \pi/3,$
$\frac{2\pi}{\lambda}\Delta x = \pi/3 \Rightarrow \Delta x = \lambda/6$
$\frac{y_1 d}{D} = \lambda/6 \Rightarrow y_1 = \frac{\lambda D}{6d} = 0.1$ mm
distance between these positions on both sides of centre of the screen
Δy = 0.2 mm

42) $\frac{I_B}{I_D} = 9$
$\Rightarrow \left(\frac{\sqrt{I_1}+\sqrt{I_2}}{\sqrt{I_1}-\sqrt{I_2}}\right)^2 = 9$
$\Rightarrow \frac{\sqrt{I_1}+\sqrt{I_2}}{\sqrt{I_1}-\sqrt{I_2}} = 3$
$\Rightarrow \sqrt{I_1}+\sqrt{I_2} = 3\sqrt{I_1}-3\sqrt{I_2}$
$\Rightarrow 4\sqrt{I_2} = 2\sqrt{I_1}$
$\Rightarrow 2\sqrt{I_2} = \sqrt{I_1}$
$\Rightarrow \frac{I_1}{I_2} = 4$
and $\frac{a_1^2}{a_2^2} = 4$
$\frac{a_1}{a_2} = 2$

44) $\left(\frac{10\lambda D}{d}\right)_{Int.} = \left(\frac{2\lambda D}{b}\right)_{diff.}$
$\frac{10}{1mm} = \frac{2}{b}$

b = 0.2 mm

45) $\frac{a_1}{a_2} = \frac{9}{\sqrt{5}} \Rightarrow I \propto a^2$

$\frac{I_1}{I_2} = \frac{81}{5} = \frac{w_1}{w_2}$

CHEMISTRY

51) $[CH_4] = \frac{16}{16} \times \frac{1}{10} = 0.1$ M

$[H_2O] = 1$ (Liquid)

$[Cl_2] = \frac{71}{71} \times \frac{1}{10} = 0.1$ M

$[CO_2] = \frac{44}{44} \times \frac{1}{10} = 0.1$ M

52) K depends only on temperature

54) $K_p = P_{O_2}$

57)

$K_p = K_c (RT)^{\Delta ng}$

If $\Delta ng = 0$ $K_p = K_c$

59)

NCERT XI, Part-I, Pg # 193

64) $PCl_5 \rightleftharpoons PCl_3 + Cl_2$

t = 0 4 mol 0 0

At eq^m 4 – 2 2 2

Total moles = 6

67)

NCERT-XI, Pg. # 207

68) When $Q_p < K_p$, the rate of forward reaction is more than rate of backward reaction.

69) Reaction quotient $Q_C = \frac{0.3 \times 0.5}{0.6}$

$Q_C = 0.25$
$\Rightarrow Q_C > K_C$
$\Rightarrow$ Reaction proceed in backward direction

70)

NCERT class XI, Part-I Pg # 203
I Edition 2007

71) NCERT 18/46

$SO_2 + \frac{1}{2} O_2 \rightleftharpoons SO_3 + \text{heat}$ (Exothermic)
favour condition for forward direction will be removal of SO_3, lowering the temperature, lowering the volume and raising the pressure

72) (A)→(II), (B)→(I), C→(III), (D)→(IV)

73) NCERT, Pg. # 202

77) $\alpha = \frac{D_T - D_0}{D_0} = \frac{8.5 - 6}{6} = \frac{2.5}{6} = 41.66$

78)

NCERT XI, Part-I, Pg # 205

81)

* On increasing pressure, reaction shifts towards less no. of moles of gas.
* On increasing volume, reaction shifts towards more no. of moles of gas.
* When $\Delta ng = 0$, No effect on reaction.
* On increasing concentration of reactant, Reaction shift in forward direction.

83)

A	+	B	$\rightleftharpoons$	C	+	D
a		a		0		0
(a – x)		(a – x)		x		x

$x = 2(a - x)$

$\therefore x = \frac{2a}{3}$

$\therefore [A] = \frac{a}{3} = [B]$ and $[C] = [D] = \frac{2a}{3}$

$\therefore K_c = \frac{\frac{2a}{3} \times \frac{2a}{3}}{\frac{a}{3} \times \frac{a}{3}} = 4$

84)

$H_2 + I_2 \rightleftharpoons 2\ HI$

ni	1	1	0
n_{eq}	0.2	0.2	1.6

$K_c = \frac{(1.6)^2}{0.2 \times 0.2} = 64$

89) $P_{total} = 3P$

$\Rightarrow \quad P = \frac{0.318}{3} = 0.106$

$\therefore \quad K_p = 4P^3 = 4.76 \times 10^{-3}$

BIOLOGY

91)

NCERT-XII, Pg # 19,20,21

92)

NCERT-XII, Pg # 5

93)

NCERT XII, Pg. # 9, 10

96)

NCERT-XII, Pg # 11

97) NCERT (XII) Pg. # 7

98) NCERT XII Page No. # 7, 8

99)

NCERT-XII, Pg # 6,7,9,10

100)

NCERT-XII Pg.# 16,18

101)

NCERT-XII, Pg # 5,7,21

102) NCERT (XII) Pg. # 12, 13

103) NCERT (XII) Pg. # 20

105)

NCERT-XII, Pg # 12

106) NCERT-XII, Pg. # 05

107) NCERT XII, Page # 10,11

108) NCERT XII, Page No. # 20

109)

NCERT XII Pg. 14

110) NCERT (XII) Pg. # 15

111) NCERT-XII, Pg # 09

112) NCERT XII Eng Pg # 06

114)

NCERT-XII, Pg # 19

115)

NCERT-XII, Pg # 18

116) NCERT(XII) Pg # 11

117)

NCERT-XII, Pg # 12

118)

NCERT-XII, Pg # 03

119)

NCERT-XII, Pg # 08

120)

NCERT-XII, Pg # 18

121)

NCERT-XII, Pg # 22

122)

NCERT-XII, Pg # 22

123) NCERT XII # 22,23

124)

NCERT XII, Page # 13, 14, Figure 1.11

125) NCERT (XII) Pg. # 21, fig. 1.15(a)

126) NCERT-XII Pg. # 06

127) NCERT (XIIth) Pg. # 07, Fig. 1.5(b)

128)

NCERT-XII Pg. # 13

129)

NCERT XII Page # 14, Fig. # 1.11(a)

130) NCERT XII Page # 06, Fig # 1.3

131) NCERT XII Pg. # 20

132) NCERT XII Pg. # 16

133) NCERT XII Pg. No. # 07, 08

134) NCERT XII, Pg. # 3, 4

135) NCERT XII Pg.#5

136) NCERT XI Pg # 60, 61

137) NCERT (XIth) Pg. # 64

139) NCERT XI Pg. # 60

140) NCERT XI Pg.# 62, 65

141)

NCERT-XI Pg. # 60, 61

142) NCERT (XIth) Pg. # 60

143) NCERT XI Pg. # 65

144) NCERT (XIth) Pg. # 62, 63

145)

NCERT-XI, Pg # 65, 66

146)

NCERT-XI, Pg # 68, 69

147) NCERT XI Pg. # 63, 64

149) NCERT XI Pg.# 60, 61

150) NCERT-XI, Pg. # 69

151)

NCERT-XI, Pg # 61

152)

NCERT-XI, Pg # 62

154) NCERT XI Pg # 60

155) NCERT-XI Page No. 61

156)

NCERT-XI, Pg # 62, 63

157) NCERT (XI) Pg. # 65

158) NCERT (XI) Pg. # 64

159)

NCERT XI Pg. # 65

160)

NCERT-XI, Pg # 69

162)

NCERT-XI, Pg # 63

163)

NCERT-XI, Pg # 64

164)

NCERT-XI, Pg # 64

165)

NCERT-XII, Pg. # 65

166) NCERT XI Pg.# 62

167) NCERT XIth Pg.#65

168)

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NCERT-XI, Pg # 68, 69

170)

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171)

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172)

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173)

NCERT-XI, Pg # 63

174)

NCERT-XI, Pg # 64

176) NCERT XI Pg. # 68, 69

177) NCERT XI Pg. # 65

178) NCERT XI, Pg. # 68, 69

179)

NCERT-XI, Pg. # 64